PNAS RESEARCH ARTICLE | ENGINEERING

Tantalum alloy–based resonators for quantum information systems

Chen Yang[a,1], Faranak Bahrami[a,1], Guangming Cheng[b], Mayer Feldman[a], Tiffany Liu[a], Nana Shumiya[a], Nan Yao[b], Stephen A. Lyon[a], Andrew A. Houck[a], Nathalie P. de Leon[a], and Robert J. Cava[c,2]

Affiliations are included on p. 6.

Contributed by Robert J. Cava; received August 6, 2025; accepted December 22, 2025; reviewed by Amir Safavi-Naeini and Chen Wang

Utilizing tantalum (Ta) in superconducting circuits has led to significant improvements, such as high qubit lifetime (T_1) and quality factors in both qubits and resonators, suggesting that material optimization plays an important role in the development of superconducting circuits. Thus we here explore superconducting gap engineering in Ta-based devices as a powerful strategy for expanding the range of suitable host materials. By alloying 20 atomic percent (at.%) hafnium into Ta thin films, we achieve a superconducting transition temperature (T_c) of 6.09 K as observed in direct current (DC) transport measurements, reflecting an increase in the superconducting gap. We systematically vary deposition conditions to control film orientation and transport properties of Ta-Hf alloy thin films. We then confirm the enhancement in T_c via microwave measurements at millikelvin temperatures. We verify the ∼40% increase in T_c relative to bare Ta devices, while the loss contributions from two-level systems and quasi-particles remain unchanged in the low temperature regime. These findings emphasize the promise of material engineering in superconducting circuits and point to many potential material candidates for further exploration.

superconducting gap engineering | superconducting qubit | Ta-Hf alloy thin films

Significance

Superconducting circuits are foundational to quantum computing, and optimizing their materials is crucial for advancing device performance. This work demonstrates that alloying tantalum with hafnium significantly enhances the superconducting gap, up to 40% compared to pure tantalum, without increasing microwave loss. By engineering the alloy composition and film deposition, the study achieves both improved superconducting properties and compatibility with fabrication processes. These findings show that material doping and alloying can strategically expand the suite of viable materials for superconducting quantum devices. This approach provides a pathway for tailoring material properties to support the next generation of quantum technologies and offers broad relevance to researchers in materials science, quantum information, and device engineering.

Superconducting qubits are among the most advanced platforms for quantum information, offering potential scalability and compatibility with standard device fabrication techniques (1, 2). These systems have enabled key advances across a broad range of applications, including quantum computation (3–6), quantum simulation (7–11), and quantum error correction (12–15). Among the materials investigated for superconducting qubits, Ta has emerged as a compelling candidate due to its self-terminating stoichiometric surface oxide and chemical resistance to harsh postfabrication processes (16–21). The surface forms a chemically stable stoichiometric Ta_2O_5 layer and is amenable to controlled thinning through surface treatments (22, 23). These advantages allow body-centered cubic (BCC) α-phase Ta to exhibit improved microwave performance under material optimization, and further advances in Ta-based qubit performance have indeed been closely tied to ongoing progress in material engineering (17, 24–26). Buffered oxide etch (BOE) removes surface oxides and increases the quality factor, Q_{int} (23), and noble-metal encapsulation deposited immediately after sputtering suppresses the stoichiometric oxide in Ta thin films (27, 28). Recently, efforts to mitigate surface loss by treating the surface oxides and utilizing high-resistivity silicon substrates have led to Ta-based qubits with lifetime and coherence times exceeding 1 ms (17, 22, 23). Guided by these successes in reducing dielectric loss in Ta-based devices, we aim to explore material engineering approaches that enhance the superconducting gap.

Alloying Ta with another element to enhance the superconducting performance expands the range of viable material candidates available to tackle different loss channels in superconducting circuits. Prior efforts have explored alloying Ta with elements such as nitrogen (N) and hafnium (Hf). Tantalum nitride (TaN) has attracted interest since it potentially leads to an oxygen-free surface (27, 29–39). More recently, bulk studies have shown that doping Ta with 20 at.% Hf can raise the T_c to approximately 6.3 K. However, the growth and properties of Ta-Hf thin films in the context of superconducting qubits remain unexplored (40).

[1]C.Y and F.B. contributed equally to this work.

[2]To whom correspondence may be addressed. Email: rcava@princeton.edu.

This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2519940123/-/DCSupplemental.

Published January 29, 2026.

Here, we present the superconducting gap enhancement relatively to both Ta and Hf while preserving the BCC phase of Ta, thereby increasing superconducting gap without compromising cubic structure in TaHf alloy thin film (16, 40, 41). We introduce this strategy for gap engineering in superconducting devices as an effective strategy, dual-metal alloying, to understand losses from nonequilibrium QPs , which are one of dominant decoherence mechanisms in superconducting resonators (23, 42–46). Additionally, Hf's $5d^2$ configuration makes it more electropositive, favoring the formation of fewer intermediate suboxides besides stoichiometric oxide (47). Although elemental Hf is chemically less robust, alloying it into Ta produces stable films suitable for fabrication.

Our structural analysis using X-ray diffraction (XRD) and scanning transmission electron microscopy (STEM) (*SI Appendix*, Fig. S4) verifies the BCC structure across all the films, and direct current (DC) transport measurements reveal a 40% enhancement in T_c relative to undoped α-Ta (T_c = 4.3 K) (48). Surface characterization, including atomic force microscopy (AFM) and X-ray photoelectron spectroscopy (XPS), was employed to assess surface morphology and native oxide chemistry. AFM (*SI Appendix*, Fig. S6) and STEM (*SI Appendix*, Fig. S4) analyses further supported the dominant film orientation revealed by XRD. To evaluate microwave performance, coplanar waveguide (CPW) resonators were fabricated from a representative alloy film and measured at millikelvin temperatures (*SI Appendix*, *Device Fabrication*). The resonators exhibited Q_{int} comparable to that of undoped α-Ta films, indicating that the microwave loss due to TLSs remains unchanged. Notably, a higher superconducting gap is observed during microwave measurements, as evidenced by the increased activation temperature (T_{act})—the extracted critical temperature from microwave model (23), which is in agreement with DC transport. In terms of chemical resistance, the surface oxides protect the underneath alloy against aggressive acid treatment such as piranha (2:1 sulfuric acid to hydrogen peroxide) when removing hydrocarbon contamination on the surface (23, 49), while incorporation of Hf reduced the robustness of Ta against BOE as a trade-off.

Although Ta-Hf alloy thin films still require further optimization for surface oxide treatments, the untreated Ta-Hf films exhibit an enhanced superconducting gap and microwave performance comparable to native Ta, with no additional loss due to TLSs and QPs compared to α-Ta. This study highlights the utility of alloying strategies for selectively tuning material properties relevant to superconducting circuits.

Superconducting Gap Engineering

Control over film growth parameters is essential to tailor structural and superconducting characteristics for thin films (*SI Appendix*, *Film Deposition*). In this work, Ta-Hf alloy thin films were grown via DC magnetron sputtering using a custom alloy target with a nominal composition of Ta (80 at.%) and Hf (20 at.%). We verified the resulting film composition to be approximately 83:17 Ta:Hf using cross-sectional TEM with energy-dispersive X-ray spectroscopy (EDS) accounting for the different sputtering yields of Ta and Hf. More importantly, the bulk film exhibits a good homogeneity in TEM/EDS (*SI Appendix*, Fig. S10*C*). The surface composition fitted by XPS show a ratio of 78:22. The discrepancy between the surface chemistry ratios from TEM/EDS and XPS is attributed to redeposition during the TEM sample preparation. The alloy film growth rate is 0.25 nm/s, and the deposition duration is used to control both thickness and crystalline orientation. The film thicknesses for 1 ks and 2 ks depositions are estimated to be 250 nm and 470 nm, respectively. Furthermore, extended deposition duration leads to a predominantly (110) orientation in the films (Table 1). The electronic transport quality of the films is evaluated using the residual resistivity ratio (RRR), defined as the resistivity at 300 K divided by that at 6 K. RRR quantitatively reflects lattice disorder; a higher RRR value corresponds to fewer defects such as smaller grain sizes in the film. Compared to clean-limit α-Ta films, which exhibit an average RRR of $\sim$49 and dirty-limit α-Ta films with an average RRR of $\sim$9.7, our Ta-Hf films fall into the dirty-limit regime with 2 ks films show higher density of defects (48). The reduction in RRR of alloy films compared to clean-limit Ta films is attributed to doping disorder and lattice mismatch introduced by Hf doping rather than impurities or β-Ta inclusions. Our structural analysis shows that films with higher RRR values tend to adopt an out-of-plane (222) orientation, corresponding to a dominant (222) reflection in the XRD pattern due to the BCC crystal symmetry, while films with smaller RRR values display mixed (110) and (222) orientations or are dominated by (110) plane (Fig. 1 *B* and *D* and Table 1).

Similar to the correlation between the thin film orientation and the RRR values from DC transport measurements, the structural evolution of Ta-Hf thin films also influences their T_c. Across all samples, T_c values range between 6.09 ± 0.13 K (Table 2). Films exhibit slightly lower T_c as the deposition temperature is increasing. XRD analysis revealed that the full width at half maximum (FWHM) of the (222) peak decreased from 550 °C to 750 °C, indicating better crystallinity. However, at 850 °C, the structural peak broadened, suggesting a degradation in structural quality at higher temperatures. This trend suggests 750 °C as the optimum deposition temperature for achieving high crystallinity, supported by the DC transport results (Table 2). These observations are mostly valid for films deposited over 1,000 s (1 ks films), while the trend for films deposited over 2,000 s (2 ks films) suggests a similar level of disorder for different deposition temperatures.

Table 1. Fitted parameters of XRD peaks belonging to Ta-Hf alloy 1 ks thin films in Fig. 1*A*

Sputtering	(110) Peak			(222) Peak			
temp. (°C)	Location (°)	FWHM (°)	Normalized intensity	Location (°)	FWHM (°)	Normalized intensity	Lattice constant (Å)
550	37.95	0.371	0.307 ± 0.007	106.91	0.851 ± 0.015	0.15	3.321
650	38.36	0.415	0.328 ± 0.005	107.00	0.731 ± 0.016	0.49	3.324
750	N/A	N/A	N/A	107.08	0.725 ± 0.017	0.50	3.315
850	38.40	0.814	0.589 ± 0.007	N/A	N/A	N/A	3.304

Values are relative to the substrate peak at $2\theta \approx 90°$.

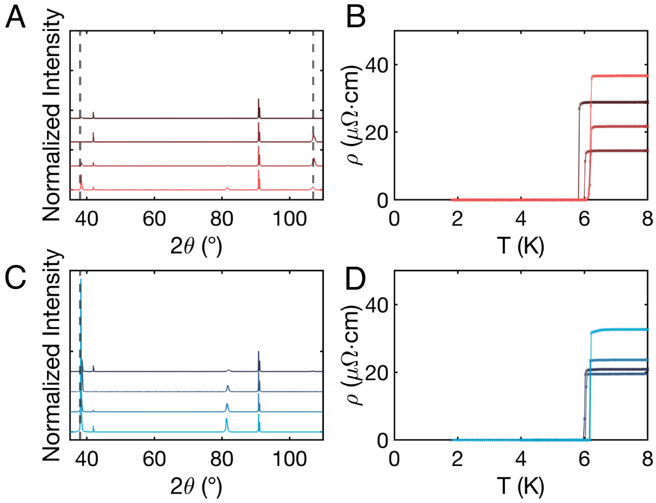

Fig. 1. Structural and DC transport characterizations. (*A*) XRD patterns of Ta-Hf alloy thin films sputtered at varying deposition temperatures and coating durations; (red tones) shows data from films coated for 1,000 s (1 ks films). (*B*) Zero-field cooling DC resistivity of the films in (*A*) plotted as a function of temperature under zero applied magnetic field; color mapping matches the corresponding deposition conditions and durations used in (*A*), illustrating how Tc and residual resistivity vary with growth conditions. (*C*) XRD patterns (blue tones) corresponds to films coated for 2,000 s (2 ks films). Within each XRD panel, the darkest to lightest traces represent films deposited at 850 °C, 750 °C, 650 °C, and 550 °C, respectively. The dashed vertical lines indicate the expected peak positions for the (110) and (222) orientations, serving as a reference for identifying crystallographic orientation. A zoomed-in plot of the (110) and (222) peaks of (*A* and *C*) are presented in *SI Appendix*, Fig. S1. The XRD patterns for each deposition are shifted along the Y-axis for clear display. (*D*) Zero-field cooling DC resistivity of the films in (*C*) plotted as a function of temperature under zero applied magnetic field. Color mapping in (*B* and *D*) matches the corresponding deposition conditions and durations used in (*A* and *C*), illustrating how T_c and residual resistivity vary with growth conditions.

The interplay between structural variation and superconducting properties in Ta-Hf films is further influenced by lattice disorder and lattice mismatching. The lattice disorder within Ta-Hf alloy thin films is a result of the Hf doping. Hf has a larger atomic radius (∼159 pm) compared to Ta (∼149 pm), thus Hf doping expands the crystal lattice for all samples, which is confirmed by peaks shifting to lower 2θ angles in XRD (Fig. 1 *A* and *C* and Table 1). This intrinsic mismatch brought the residual resistivity (ρ ($\mu\Omega\cdot$cm)) of the Ta-Hf alloy thin films to nearly five times that of the dirty-limit Ta films (Fig. 1 *B* and *D* and Table 2) (48). Furthermore, deposition conditions lead to lattice mismatch between the film and the substrate. XRD and STEM analyses reveal that the average lattice mismatch between the Ta-Hf alloy thin films and the sapphire substrate is approximately 1.47% for the 1 ks films and 1.49% for the 2 ks films (Table 2). The 2 ks film we selected for resonators measurements has a lattice mismatch of 1.94%, which is similar to the 1.9% mismatch of dirty-limit Ta films. On the other hand, the 1 ks film selected has a mismatch of 1.50% which is similar to clean-limit Ta films, suggesting the 1 ks film host fewer defects compared to 2 ks film (48). Both lattice disorder and lattice mismatch contribute to increased residual resistivity and decreased RRR, emphasizing the critical role of lattice compatibility in optimizing transport properties (39).

Parameterizing Microwave Losses

In superconducting microwave resonators, energy dissipation arises primarily from TLSs, QPs, trapped vortices, and power- and temperature-independent loss (Q_{other}) (23, 48). TLSs are microscopic defects, located in amorphous dielectrics or at material interfaces in the resonators. These defects are particularly influential at low temperatures and single-photon microwave power, where they act as the dominant source of loss. The loss due to QPs becomes dominant at elevated temperatures and leads to an exponential decay in internal quality factor (Q_{int}) at higher temperatures (>0.5 K). The slope of this decay is directly influenced by the superconducting gap.

We investigate the microwave loss behavior in the Ta-Hf thin films by measuring CPW resonators fabricated from this alloy, deposited at 750 °C for 1,000 and 2,000 s. By conducting microwave measurements across varying temperatures and microwave powers, we aim to evaluate different loss behaviors (e.g., TLSs, QPs, and vortices) compared to the bare Ta devices. Our findings reveal that at base temperature ($T \sim 20$ mK), Q_{int} increases with rising microwave power, indicative of loss from saturable TLSs. At base temperature, Q_{int} spans from approximately 1×10^6 at the lowest drive power to 4×10^7 at the highest; a similar range of Q_{int} is observed in the Ta devices (Fig. 2*A*). Although the low and mid-temperature range shows similar behavior for both Ta-Hf and Ta devices (23), the activation temperature due to QPs is demonstrably increased in Ta-Hf resonators, suggesting that the superconducting gap is enhanced. Across seven measured resonators, the extracted T_{act} for Ta-Hf resonators exhibit a mean-value of 5.92 ± 0.24 K for 2 ks film, consistent with DC transport measurements (Fig. 1*D* and Table 2). For the 1 ks film resonators, the extracted T_{act} of 4.40 ± 0.46 K is lower than the value obtained from DC transport measurements (6.09 ± 0.13 K) (Fig. 1*B*). This discrepancy is attributed to vortex-motion-induced losses, which are more apparent in the 1 ks film due to its relatively

Table 2. Fitted parameters for resistivity measurements of Ta-Hf alloy thin films with different coating times and sputtering temperatures

Coating time (s)	Sputtering temperature (°C)	T_c (K)	RRR	Residual resistivity ($\mu\Omega\cdot$cm)	Lattice mismatch%
1,000	550	6.17 ± 0.01	1.55 ± 0.01	36.61 ± 0.01	1.31
	650	6.20 ± 0.01	2.31 ± 0.06	21.72 ± 0.37	1.22
	750	6.02 ± 0.01	1.91 ± 0.01	14.48 ± 0.01	1.50
	850	5.84 ± 0.01	1.75 ± 0.01	28.85 ± 0.08	1.85
2,000	550	6.21 ± 0.01	1.67 ± 0.01	32.55 ± 0.01	1.10
	650	6.18 ± 0.01	1.59 ± 0.01	23.64 ± 0.01	1.26
	750	6.06 ± 0.01	1.83 ± 0.01	20.88 ± 0.01	1.94
	850	6.00 ± 0.01	2.06 ± 0.01	20.89 ± 0.01	1.67

Residual resistivity (ρ) and RRR are estimated from full-range ρ vs. T measurements. The error in T_c is defined as the data point collection increment measured by PPMS.

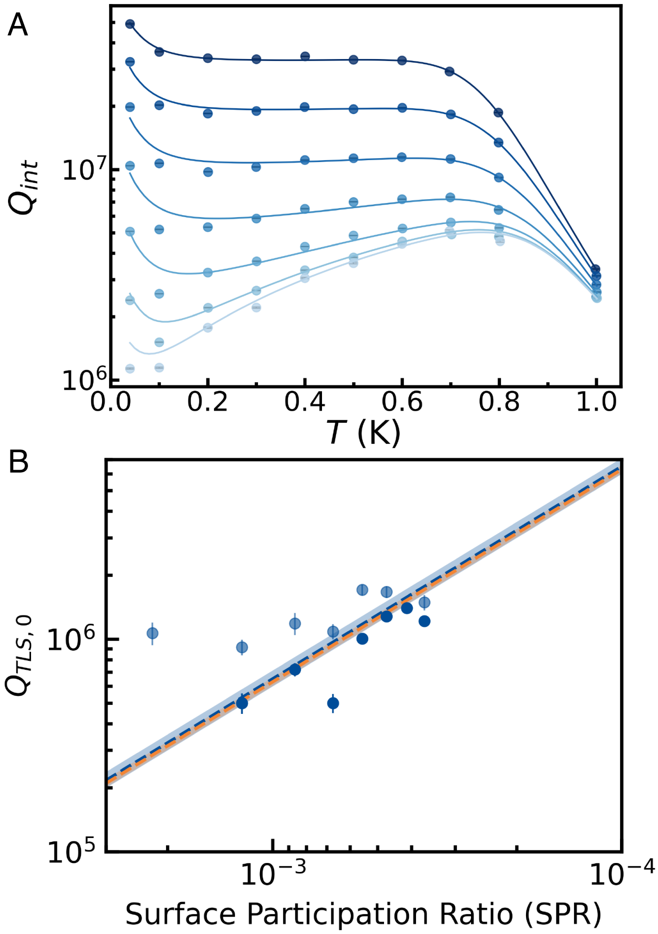

Fig. 2. Microwave losses in Ta-Hf resonators. (*A*) Temperature dependence of the internal quality factor (Q_{int}) measured at various microwave powers for a CPW resonator with gap width of 10 μm made of 2 ks Ta-Hf film. At low temperatures and low powers, Q_{int} is limited by saturable TLSs and rises with increasing power. At higher temperatures, an exponential decay in Q_{int} indicates increasing losses from thermally activated QPs. The extracted T_{act} for Ta-Hf 1 ks and 2 ks resonators show mean-values of 4.40 ± 0.46 and 5.92 ± 0.24 K, respectively. They confirm the enhancement in the superconducting gap. The design used for CPW device is identical to ref. 23. (*B*) Extracted $Q_{\mathrm{TLS},0}$ for fourteen Ta-Hf resonators plotted as a function of surface participation ratio (p_{MS}) (darker blue for 2 ks resonators and lighter blue for 1 ks resonators). The extracted surface loss tangent for Ta-Hf (blue dashed line) and Ta (orange dashed line, ref. 23) are $(15.4 \pm 1.3) \times 10^{-4}$ and $(15.9 \pm 0.7) \times 10^{-4}$, respectively. The shaded areas correspond to the uncertainties in the surface loss tangents.

lower lattice mismatch and larger grain sizes. The structural characteristics revealed by SEM and AFM analyses indicate larger grain sizes (*SI Appendix*, Fig. S10), thereby enabling more vortex motion during microwave measurements compared to 2 ks film resonators. Similar behavior is observed for Ta films with different defect densities (48). We confirm that the loss due to the QPs remains comparable to the Ta devices while the T_c is increased by 40% (*SI Appendix*, Fig. S5) (23, 48). To quantitatively assess TLS-induced losses in the Ta-Hf films, we extracted the inverse linear absorption from TLSs, ($Q_{\mathrm{TLS},0}$), and extracted the surface loss for measured Ta-Hf resonators. The combined surface loss tangent ($\tan\delta$) for the Ta-Hf alloy is $(15.4 \pm 1.3) \times 10^{-4}$, which within the uncertainty overlaps the surface loss tangent of $(15.9 \pm 0.7) \times 10^{-4}$ measured for native Ta (Fig. 2*B*) (23). The trend agrees well with reference data from native Ta resonators (23), indicating that Hf incorporation does

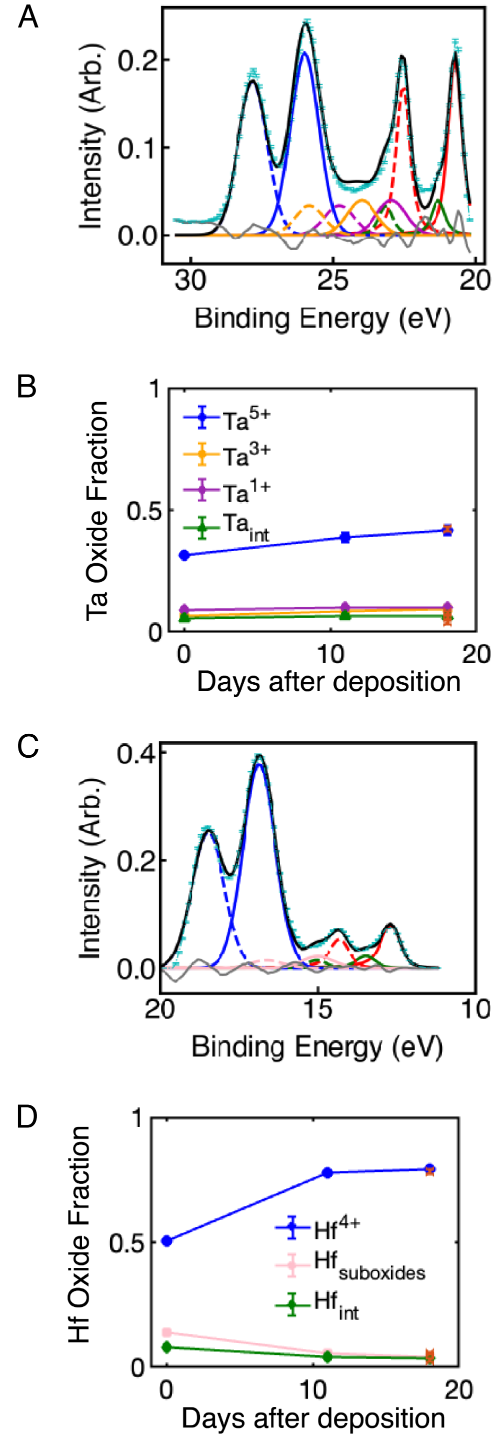

Fig. 3. Surface oxide evaluation over 18 d of exposure to ambient. (*A*) Ta binding energy spectra from XPS measurements. (*B*) Atomic percentages of each Ta oxide species and the interface layer for samples measured over 18 d. (*C*) Hf binding energy spectra from XPS measurements. (*D*) Atomic percentages of each Hf oxide species for samples measured over 18 d. Both metallic and oxidation states exhibit a pair of peaks due to spin-orbit splitting. The metallic peaks show a sharp asymmetric shape, while the oxide peaks are modeled with Gaussian profile doublets. The metallic doublet peaks (red) are located at 11 to 14 eV and 20 to 24 eV for Hf and Ta, respectively. The x-axis in (*B* and *D*) refers to the time variation from the deposition date. The uncertainties are the SD of the fit. The legend in panel (*B*) corresponds to both (*A* and *B*), and the legend in panel (*D*) corresponds to both (*C* and *D*). The oxide fraction trace for the 1 ks alloy film resonators treated with piranha solution is shown in panels (*B* and *D*) as orange cross markers. The close overlap indicates that piranha treatment does not alter the surface chemistry. The resonators are fabricated under conditions of oxide saturation of both Ta_2O_5 and HfO_2. The marker sizes are bigger than error bars. We present the XPS spectra of day 0 and 18 for Ta and Hf in *SI Appendix*, Fig. S8.

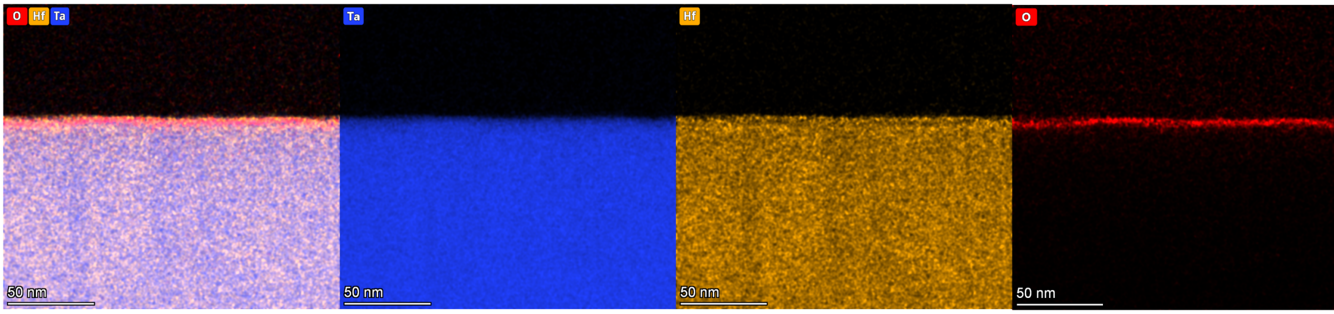

Fig. 4. Cross-sectional EDS elemental mapping on Ta-Hf. Cross-sectional EDS mapping Ta (blue) and Hf (orange) confirmed a uniform Ta-Hf alloy with the Ta to Hf atomic ratios for the bulk to be (83% : 17%). The oxygen (red) map reveals a surface oxide layer with the estimated surface oxide thickness of around 4 nm. The 1 ks Ta-Hf film exhibits an oxide thickness of approximately 3 nm, similar to that of native Ta (*SI Appendix*, Fig. S10). The EDS resolution limit is 0.5 atomic percent.

not introduce additional TLS loss and suggesting that alloying with Hf preserves dielectric performance.

These results confirm that the dominant low-temperature dissipation in the Ta-Hf alloy resonators originates from saturable TLSs, and that Hf doping does not compromise the microwave performance at conditions relevant to qubit performance. However, its weak chemical resistance to BOE limits further improvements in the surface oxide treatments for these devices (*SI Appendix*, Fig. S7).

Surface Oxide Chemistry in the Ta-Hf Alloy

To further investigate the dielectric loss limiting the performance of Ta-Hf alloys relative to native Ta, we employ XPS to characterize the surface oxide in our Ta-Hf thin films (*SI Appendix*, *XPS Analysis*). Core-level spectra were collected for the Ta and Hf 4*f* electrons to identify the chemical states present on the film surface (Fig. 3). For Ta, the metallic doublets appear in the 20 to 24 eV range and the Ta^{5+} doublets are observed between 25 to 29 eV, which align with those reported for native Ta (Fig. 3*A*) (22, 50, 51). Suboxides of Ta, specifically $Ta^{1+}_{7/2}$ and $Ta^{3+}_{7/2}$, are fitted at 23 eV and 24 eV, respectively, which fall within the reported suboxides' peak range in native Ta. Hf oxides are analyzed similarly, although distinguishing individual suboxides is more challenging. The energy separation between the 4*f* peaks for common Hf suboxides, such as Hf^{2+} and Hf^{3+}, is less than 0.5 eV, making them difficult to resolve given the typical instrumental resolution of 0.1 to 0.2 eV. Therefore, we refined the Hf spectra using broader peaks attributed to fully oxidized HfO_2 and a less-defined HfO_x state, following established Hf XPS analyses (Fig. 3*C*) (52, 53). Reported literature finds the $Hf^{4+}_{7/2}$ peak between 16 to 18 eV, while HfO_x peaks appear around 14 to 16 eV, depending on the local chemical environment, with spin–orbit splitting values typically around 1.7 ± 0.3 eV. Our fitted $Hf^{4f_{7/2}}$ between 16 to 18 eV align well with these reported values, as does the range for $Hf^{suboxides}_{7/2}$ between 14 to 16 eV (54) (Fig. 3*C*). Additional spectral shifts are observed due to different material interfaces. The Ta metal-oxide interface causes a shift in the Ta metallic doublets by approximately 0.4 eV, consistent with prior reports for native Ta films (22). For Hf, a similar metal-oxide interfacial component (Hf^{int}) is shifted by 0.7 eV relative to the bulk metal peak (52, 54, 55).

We monitored oxide growth over an 18-d period following film deposition to better understand how Hf incorporation influences surface oxides. In both Ta and Hf spectra, the fraction of fully oxidized (Ta_2O_5 and HfO_2) states gradually increased during the 18-d exposure, indicating progressive surface oxidation (Fig. 3 *B* and *D*). To estimate the surface oxide thickness, we use cross-sectional TEM/EDS mapping (Fig. 4 and *SI Appendix*, Fig. S10). The analysis reveals that the oxide thickness for the 1 ks Ta-Hf alloy film is approximately 3 nm thick (*SI Appendix*, Fig. S10), comparable to the ∼3 nm oxide observed on native Ta (22). For the 2 ks Ta-Hf film, the oxide thickness increases slightly to about 4 nm (Fig. 4). The measured surface loss tangent of the Ta-Hf alloy remains comparable to that of native Ta. This result suggests that the surface oxide of the Ta-Hf alloy, which contains a measurable fraction of HfO_x, does not introduce additional dielectric loss.

Conclusion

We demonstrate that the superconducting gap of Ta-based thin films can successfully be increased through alloying with Hf, achieving higher T_c (6.09±0.13) without introducing additional microwave losses attributable to TLSs or QPs. Despite the alloy has a fraction of HfO_x and increased surface roughness in the sidewalls of resonators, the microwave loss remains comparable to native Ta devices, suggesting that the Hf oxides have similar loss behavior as Ta oxides. Furthermore, Ta-Hf alloys retain both their superconductivity and microwave performance while withstanding aggressive acid treatments that typically damage pure Hf films. By tuning the superconducting gap, this alloying strategy enables suppression of QP-related losses at elevated temperatures. These findings position dual-metal alloying as a promising and underexplored pathway for designing materials that target specific loss mechanisms in superconducting circuits. Beyond identifying Hf as a useful dopant, we propose a broader materials strategy: Elements with low intrinsic T_c or poor standalone compatibility with fabrication processes may still offer desirable properties, such as favorable oxide chemistry or higher gap energy when incorporated into an alloy. Future efforts may explore other binary or ternary systems to further tailor specific material characteristics. Last, this improvement can be implemented into the high frequency/high temperature transmon qubits (44) that already operate in a temperature regime more strongly affected by the quasi-particle loss or gap engineered devices (45) where variation in the thickness is used for superconducting gap engineering.

Data, Materials, and Software Availability. Excel, csv, matlab, python, and dat data have been deposited in Harvard Dataverse (DOI will be provided upon publication). All other data are included in the manuscript and/or *SI Appendix*.

ACKNOWLEDGMENTS. This material is based upon work supported by the U.S. Department of Energy, Office of Science, National Quantum Information Science Research Centers, Co-design Center for Quantum Advantage (C^2QA) under contract No. DE-SC0012704. We acknowledge the use of Princeton's Imaging and Analysis Center, which is partially supported by the Princeton Center for Complex Materials, a NSF–MRSEC program (DMR-2011750), as well as Princeton Mirco/Nano Fabrication Laboratory. Princeton University Professor Andrew Houck is also a consultant for Quantum Circuits Incorporated (QCI). Due to his income from QCI, Princeton University has a management plan in place to mitigate a potential conflict of interest that could affect the design, conduct, and reporting of this research.

Author affiliations: [a]Department of Electrical and Computer Engineering, Princeton University, Princeton, NJ 08540; [b]Princeton Materials Institute, Princeton University, Princeton, NJ 08540; and [c]Department of Chemistry, Princeton University, Princeton, NJ 08540

Author contributions: N.P.d.L. and R.J.C. designed research; C.Y., F.B., and G.C. performed research; M.F., T.L., N.S., N.Y., S.A.L., and A.A.H. contributed new reagents/analytic tools; C.Y. and F.B. analyzed data; N.Y., S.A.L., A.A.H., N.P.d.L., and R.J.C. contributed conceptual consultation, valuable discussions, and instruments essential for this work; and C.Y., F.B., N.P.d.L., and R.J.C. wrote the paper.

Reviewers: A.S.-N., Stanford University; and C.W., University of Massachusetts-Amherst.

Competing interest statement: N.P.d.L., Princeton University professor, is a visiting faculty researcher with Google Quantum AI. As a result of her income from Google, Princeton University has a management plan in place to mitigate a potential conflict of interest that could affect the design, conduct and reporting of this research. Her academic group also has a sponsored research contract with Google Quantum AI.

1. M. Kjaergaard *et al*., Superconducting qubits: Current state of play. *Annu. Rev. Condens. Matter Phys.* **11**, 369–395 (2020).
2. A. Blais, A. L. Grimsmo, S. M. Girvin, A. Wallraff, Circuit quantum electrodynamics. *Rev. Mod. Phys.* **93**, 025005 (2021).
3. F. Arute *et al*., Quantum supremacy using a programmable superconducting processor. *Nature* **574**, 505–510 (2019).
4. Y. Sung *et al*., Realization of high-fidelity CZ and ZZ-free iSWAP gates with a tunable coupler. *Phys. Rev. X* **11**, 021058 (2021).
5. P. Krantz *et al*., A quantum engineer's guide to superconducting qubits. *Appl. Phys. Rev.* **6**, 021318 (2019).
6. Z. Chen *et al*., Exponential suppression of bit or phase errors with cyclic error correction. *Nature* **595**, 383–387 (2021).
7. P. Roushan *et al*., Spectroscopic signatures of localization with interacting photons in superconducting qubits. *Science* **358**, 1175–1179 (2017).
8. A. Kandala *et al*., Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **549**, 242–246 (2017).
9. X. Mi *et al*., Time-crystalline eigenstate order on a quantum processor. *Nature* **601**, 531–536 (2022).
10. Y. Salathé *et al*., Digital quantum simulation of spin models with circuit quantum electrodynamics. *Phys. Rev. X* **5**, 021027 (2015).
11. J. Zhang *et al*., Observation of a many-body dynamical phase transition with a 53-qubit quantum simulator. *Nature* **551**, 601–604 (2017).
12. R. Acharya *et al*., Quantum error correction below the surface code threshold. *Nature* **638**, 920–926 (2025).
13. C. K. Andersen *et al*., Repeated quantum error detection in a surface code. *Nat. Phys.* **16**, 875–880 (2020).
14. S. Krinner *et al*., Realizing repeated quantum error correction in a distance-three surface code. *Nature* **605**, 669–674 (2022).
15. S. Bravyi, D. Gosset, R. König, Quantum advantage with shallow circuits. *Science* **362**, 308–311 (2018).
16. A. P. M. Place *et al*., New material platform for superconducting transmon qubits with coherence times exceeding 0.3 milliseconds. *Nat. Commun.* **12**, 1779 (2021).
17. M. P. Bland *et al*., Millisecond lifetimes and coherence times in 2D transmon qubits. *Nature* **647**, 343–348 (2025).
18. S. Ganjam *et al*., Surpassing millisecond coherence in on chip superconducting quantum memories by optimizing materials and circuit design. *Nat. Commun.* **15**, 3687 (2024).
19. C. Wang *et al*., Towards practical quantum computers: Transmon qubit with a lifetime approaching 0.5 milliseconds. *NPJ Quantum Inf.* **8**, 1–6 (2022).
20. V. Sivak *et al*., Real-time quantum error correction beyond break-even. *Nature* **616**, 50–55 (2023).
21. D. Gao *et al*., Establishing a new benchmark in quantum computational advantage with 105-qubit zuchongzhi 3.0 processor. arXiv [Preprint] (2024). http://arxiv.org/abs/2412.11924 (Accessed 16 April 2025).
22. R. A. McLellan *et al*., Chemical profiles of the oxides on tantalum in state of the art superconducting circuits. *Adv. Sci.* **10**, 2300921 (2023).
23. K. D. Crowley *et al*., Disentangling losses in tantalum superconducting circuits. *Phys. Rev. X* **13**, 041005 (2023).
24. N. P. De Leon *et al*., Materials challenges and opportunities for quantum computing hardware. *Science* **372**, eabb2823 (2021).
25. C. E. Murray, Material matters in superconducting qubits. *Mater. Sci. Eng. R Rep.* **146**, 100646 (2021).
26. S. E. De Graaf, S. Un, A. G. Shard, T. Lindström, Chemical and structural identification of material defects in superconducting quantum circuits. *Mater. Quantum. Technol.* **2**, 032001 (2022).
27. R. D. Chang *et al*., Eliminating surface oxides of superconducting circuits with noble metal encapsulation. *Phys. Rev. Lett.* **134**, 097001 (2025).
28. C. Zhou *et al*., Ultrathin magnesium-based coating as an efficient oxygen barrier for superconducting circuit materials. *Adv. Mater.* **36**, 2310280 (2024).
29. H. B. Nie *et al*., Structural and electrical properties of tantalum nitride thin films fabricated by using reactive radio-frequency magnetron sputtering. arXiv [Preprint] (2003). http://arxiv.org/abs/cond-mat/0305683 (Accessed 10 April 2025).
30. J. S. Oh *et al*., Structure and formation mechanisms in tantalum and niobium oxides in superconducting quantum circuits. *ACS Nano* **18**, 19732–19741 (2024).
31. V. Quintanar-Zamora *et al*., Stable oxygen incorporation in superconducting TaN: An experimental and theoretical assessment. *ACS Omega* **9**, 35069–35079 (2024).
32. M. Müller *et al*., Growth optimization of TaN for superconducting spintronics. *Mater. Quantum. Technol.* **1**, 045001 (2021).
33. K. S. Il'in, D. Rall, M. Siegel, A. Semenov, Critical current density in thin superconducting TaN film structures. *Phys. C Supercond.* **479**, 176–178 (2012).
34. M. Cedillo Rosillo, O. Contreras López, J. A. Díaz, A. Conde Gallardo, H. A. Castillo Cuero, High-temperature epitaxial growth of tantalum nitride thin films on MgO substrates: Structural evolution and potential for SQUID applications. Beilstein [Preprint] (2024). https://www.beilstein-archives.org/xiv/preprints/202461 (Accessed 21 April 2025).
35. S. Chaudhuri, I. J. Maasilta, L. Chandernagor, M. Ging, M. Lahtinen, Fabrication of superconducting tantalum nitride thin films using infrared pulsed laser deposition. *J. Vac. Sci. Technol. A* **31**, 061502 (2013).
36. K. Reichelt, W. Nellen, G. Mair, Preparation and compositional analysis of sputtered TaN films. *J. Appl. Phys.* **49**, 5284–5287 (1978).
37. S. I. Baik, Y. W. Kim, Microstructural evolution of tantalum nitride thin films synthesized by inductively coupled plasma sputtering. *Appl. Microsc.* **50**, 7 (2020).
38. M. D. Serna-Manrique *et al*., Growth mechanisms of TaN thin films produced by DC magnetron sputtering on 304 steel substrates and their influence on the corrosion resistance. *Coatings* **12**, 979 (2022).
39. M. Drimmer *et al*., The effect of niobium thin film structure on losses in superconducting circuits. arXiv [Preprint] (2024). http://arxiv.org/abs/2403.12164 (Accessed 26 June 2025).
40. T. Klimczuk, S. Królak, R. J. Cava, Superconductivity of Ta-Hf and Ta-Zr alloys: Potential alloys for use in superconducting devices. *Phys. Rev. Mater.* **7**, 064802 (2023).
41. W. Gey, D. Köhnlein, Superconductivity of Hf-Ta and Ta-W alloys under pressure. *Z. Phys. A Hadrons Nucl.* **255**, 308–314 (1972).
42. H. Nho *et al*., Recovery dynamics of a gap-engineered transmon after a quasiparticle burst. arXiv [Preprint] (2025). http://arxiv.org/abs/2505.08104 (Accessed 22 September 2025).
43. P. Kamenov, T. DiNapoli, M. Gershenson, S. Chakram, Suppression of quasiparticle poisoning in transmon qubits by gap engineering. arXiv [Preprint] (2024). http://arxiv.org/abs/2309.02655 (Accessed 22 September 2025).
44. A. Anferov, F. Wan, S. P. Harvey, J. Simon, D. I. Schuster, Millimeter-wave superconducting qubit. *PRX Quantum* **6**, 020336 (2025).
45. M. McEwen *et al*., Resisting high-energy impact events through gap engineering in superconducting qubit arrays. *Phys. Rev. Lett.* **133**, 240601 (2024).
46. V. D. Kurilovich *et al*., Correlated error bursts in a gap-engineered superconducting qubit array. arXiv [Preprint] (2025). http://arxiv.org/abs/2506.18228 (Accessed 28 October 2025).
47. R. Dronskowski, S. Kikkawa, A. Stein, *Handbook of Solid State Chemistry, 6 Volume Set* (John Wiley & Sons, 2017).
48. F. Bahrami *et al*., Vortex motion induced losses in tantalum resonators. arXiv [Preprint] (2025). http://arxiv.org/abs/2503.03168 (Accessed 7 April 2025).
49. J. L. Collins *et al*., Electrical and chemical characterizations of hafnium (IV) oxide films for biological lab-on-a-chip devices. *Thin Solid Films* **662**, 60–69 (2018).
50. D. Wolverson *et al*., First-principles estimation of core level shifts for Hf, Ta, W, and Re. *J. Phys. Chem. C* **126**, 9135–9142 (2022).
51. E. Atanassova, D. Spassov, X-ray photoelectron spectroscopy of thermal thin Ta2O5 films on Si. *Appl. Surf. Sci.* **135**, 71–82 (1998).
52. C. Morant, L. Galán, J. M. Sanz, An XPS study of the initial stages of oxidation of hafnium. *Surf. Interface Anal.* **16**, 304–308 (1990).
53. V. Sammelselg *et al*., XPS and AFM investigation of hafnium dioxide thin films prepared by atomic layer deposition on silicon. *J. J. Electron Spectrosc. Relat. Phenom.* **156–158**, 150–154 (2007).
54. D. Barreca, A. Milanov, R. A. Fischer, A. Devi, E. Tondello, Hafnium oxide thin film grown by ALD: An XPS study. *Surf. Sci. Spectra* **14**, 34–40 (2009).
55. H. Wang *et al*., Study of reactions between HfO2 and Si in thin films with precise identification of chemical states by XPS. *Appl. Surf. Sci.* **257**, 3440–3445 (2011).